A tiny switch with no moving parts

Transistors

Name _____ Date _____



Little Wonderer finds a black bead with three thin legs. It looks like a bug, but it is a transistor. Big Wonderer leans in beside the little one. This one is big enough to see, but most are far too small. Billions of them work inside the machines around us. What could something so small be doing?



Long ago, machines used glass vacuum tubes instead. A tube was about as tall as a hand. It glowed, got hot and burned out like an old light bulb. One early computer needed about 18,000 tubes and filled a whole room. In 1947, three scientists built the first transistor from gold foil and a small crystal.



Think of the tap on a sink. A small turn of the handle lets a big stream of water through. A transistor works a bit like that. Electricity flows in one leg and out another. A tiny signal on the middle leg opens or closes the path. That is how a small push can control a much bigger flow.

transistor	a tiny switch that controls a flow of electricity
vacuum tube	an old glass part used before transistors
signal	a small push that tells something what to do
silicon	a material found in sand, used to make transistors
semiconductor	a material that carries electricity only some of the time

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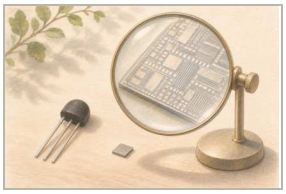
Name _____ Date _____



Most transistors are made of silicon, and silicon is found in ordinary sand. Pure silicon barely lets electricity through. Makers add a tiny sprinkle of other atoms, a little like a pinch of salt in soup. That sprinkle lets the silicon carry electricity only when it is told to. A material like this is called a semiconductor.



Little Wonderer points along a row of eight paper cards. Light side up means on, and dark side up means off. A transistor switch is either on or off, like a one or a zero. The little one flips a card and makes a new pattern. Eight switches can make 256 different patterns, enough for every letter on a keyboard.



A chip smaller than a fingernail holds billions of transistors. Light shines through a stencil to print them onto a silicon wafer, layer after layer. In many tiny transistors, the middle part never touches the path. It sits behind a thin wall and steers the flow with an unseen push. How can something unseen change where electricity goes?

1. What came before transistors?
2. How is a transistor like the tap on a sink?
3. What do makers add to silicon, and why?
4. How can a row of on-off switches make many patterns?

Your turn

Write or dictate three sentences: what a transistor is made of, what its middle leg does, and what on and off mean.

Example: A transistor is made of silicon. Its middle leg opens the path. On and off make ones and zeros.